

Peng Lin

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EDUCATION

Syracuse University

BS in Computer Science, Minor in Mathematics

- *ECS Leadership Scholar*

Expected graduation: May 2027

Syracuse, NY

GPA: 3.788

EXPERIENCE

AI Research Intern

Leibniz Lab, Syracuse University — advised by Prof. Shakarian

Spring 2026 – Present

Syracuse, NY

- Contribute to PyReason, an open-source neurosymbolic reasoning library, by authoring documentation and tutorials covering rule and fact loading from files, LLM-generated PyReason rules, and natural language to PyReason syntax translation
- Develop the LLM component of VulReasoner, an ARPA-H UPGRADE-funded vulnerability reasoning system, designing prompt pipelines that translate unstructured text into structured logical rules

Research Assistant

LinkLab, Syracuse University — advised by Prof. Banks

May 2026 – Present

Syracuse, NY

- Run weekly data collection sessions at Destiny Mall for the AI Companion team's perception studies, supporting research on how people attribute mental and moral capacities to social machines

Research Assistant — VR Development

Department of EECS, Syracuse University — advised by Prof. Ghosh

Jan 2026 – May 2026

Syracuse, NY

- Developed interactive VR simulation in Unity/C# for semiconductor cleanroom training, implementing particle systems and collision detection to visualize contamination scenarios
- Delivered production-ready executable application within one-week sprint for demonstration, collaborating with 2 faculty advisors and undergraduate/graduate research team
- Presented simulation to Q-SUCCEED-CNY program cohort — an NSF-affiliated workforce initiative preparing adult career changers and veterans for semiconductor roles, in partnership with Micron

PROJECTS

Personal Finance Dashboard | *Python, Flask, React, SQLAlchemy*

- Built full-stack web application to parse and analyze credit card transactions from multiple financial institutions (Apple Card, Bank of America), processing 900+ real transactions with automated data normalization
- Designed RESTful API with Flask handling inconsistent CSV formats from different banks, implementing data standardization pipeline to resolve conflicting sign conventions and date formats
- Developed rule-based transaction categorization engine achieving 92% accuracy across 7 spending categories, reducing uncategorized transactions from 85% to 8% through iterative keyword optimization

WatchToRead | *Python, Flask, Claude API, ReportLab*

- Built end-to-end AI content pipeline converting YouTube video transcripts into structured PDF eBooks (8–9 pages per video), integrating YouTube Transcript API and Claude API with prompt engineering to produce audience-tailored output in under 30 seconds
- Designed modular service architecture separating transcript extraction, AI processing, and PDF rendering into three single-responsibility modules, isolating LLM provider dependency to a single file
- Engineered audience-adaptive content generation across four complexity levels (beginner to expert), enabling the same transcript to produce contextually appropriate output for different readers

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, C#, SQL, HTML/CSS

Frameworks: React, Flask, Next.js, Tailwind, pandas, SQLAlchemy

Tools & Platforms: Git, SQLite, PostgreSQL, VS Code, Unity, Promptfoo, Linux

Concepts: RESTful API Design, LLM APIs, Prompt Engineering, Agile/Scrum